What makes a Steam game successful? – ML project, stage 1

Stage 1 of a machine learning project (Machine Learning course, Computer Science, Vistula University in Warsaw, winter semester 2026/2027): two public datasets about Steam games, exploratory data analysis and research questions. Code, data and reports: github.com/ProstoKery/steam-ml-project.

1. Team

Name	ID	GitHub	Scope of work
Serhiienko Kyrylo	71897	ProstoKery	dataset 1 (reviews, text data): download, cleaning and exploratory analysis; GitHub repository and README; editing the report
Zavoritnii Roman	72086	yourromanbm-dot	dataset 2 (games, numerical data): exploratory analysis and statistical tests; joining both datasets
together	–	–	choice of topic and datasets, research questions

2. Topic and data at a glance

Steam is the largest PC game store: our dataset contains more than 124 thousand games, and almost 25 thousand were released in 2025 alone. Players rate games by writing a review and marking whether they recommend the game. We want to find out what makes a game well rated and whether the rating can be predicted: from the text of reviews (dataset 1) and from the features of the game itself (dataset 2). The two datasets share the Steam game ID (`item_id` = `appid`), so we can compare what players write with the numbers that describe the game.

	Dataset 1: reviews (natural language)	Dataset 2: games (numerical data)
Content	review text + whether the player recommends the game	price, ratings, number of players, genres, platforms etc.
Size	59,305 reviews, 25,458 users, 3,682 games	124,146 games, 41 columns in the source file
Period	reviews from 2010–2016	games released 1997–2026, ratings as of the collection date (2026)
Source	UCSD, J. McAuley's group; copy on Hugging Face	FronkonGames (Steam API + SteamSpy); Hugging Face
License	not stated in the repository; the authors ask to cite their publications (section 3.1)	CC BY 4.0 according to the dataset card (repository metadata says MIT)

3. About the data

3.1 Dataset 1: Steam user reviews

- **Origin:** the Steam review and bundle dataset collected by Prof. Julian McAuley's group (University of California San Diego) from public Steam Community profiles. We use the "australian user reviews" part: reviews by users from Australia, each with its text and whether the author recommends the game.
- **Publications to cite** (requested by the authors of the dataset): W.-C. Kang, J. McAuley, *Self-attentive sequential recommendation*, ICDM 2018; M. Wan, J. McAuley, *Item recommendation on monotonic behavior chains*, RecSys 2018; A. Pathak, K. Gupta, J. McAuley, *Generating and personalizing bundle recommendations on Steam*, SIGIR 2017.
- **Where we download it from:** a copy of the dataset on Hugging Face, recommender-system/steam-review-and-bundle-dataset, file `australian_user_reviews.json.gz` (6.94 MB), commit `8e9c6b569122fb20668f5da3b0f3970a853fc5d8`, file SHA-256 `5b2add00…bf6dfc` (full value in `data/processed/sources.json`).
- **Subset:** of the five files in the repository we only use `australian_user_reviews.json.gz`. The others (users' game libraries, bundles, version 2 reviews without the "recommended" label) are not needed.
- **Preparation** (`src/01_download_data.py`): each line of the file is a dictionary in Python syntax rather than strict JSON, so we parse it with `ast.literal_eval`. We expand each user's list of reviews (one review = one row). Steam account names are replaced with consecutive numbers (anonymisation). From the texts "Posted November 5, 2011." and "2 of 3 people (67%) found this review helpful" we extract the date and the vote counts. Result: `data/processed/reviews.csv.gz`, 59,305 rows, 14 columns.

3.2 Dataset 2: Steam game metadata

- **Origin:** the "Steam Games Dataset" (FronkonGames): data from the Steam store (Steam API) complemented with SteamSpy data, collected with the open-source tool FronkonGames/Steam-Games-Scraper. The dataset covers games released on Steam (no DLC, videos or soundtracks). DOI: 10.57967/hf/0511.
- **Where we download it from:** FronkonGames/steam-games-dataset, file `data/train-00000-of-00001.parquet` (192 MB, 124,146 rows, 41 columns), commit `6a31fcb42082345acc817e176b28849b7c05f248`, SHA-256 `775b17f2…388c5a77`.
- **Subset:** we keep all games and 28 of the 41 columns. We drop 13 columns with long descriptions, URLs and media (`detailed_description`, `short_description`, `reviews`, `header_image`, `website`, `support_url`, `support_email`, `metacritic_url`, `score_rank`, `notes`, `screenshots`, `movies`, `packages`). We add 10 derived columns: release year, the lower and upper bound of the owner-count range, and the number of items in list columns (languages, developers, publishers, categories, genres, tags). Result: `data/processed/games.csv.gz`, 124,146 rows, 38 columns.

3.3 How to reproduce the data

1. `python -m pip install -r requirements.txt`

2. `python src/01_download_data.py` – downloads both files from Hugging Face (at the commits given above), prepares the subsets and saves them in `data/processed/` together with `sources.json` (repository, file, commit, SHA-256, download date). We downloaded our data on 28.10.2026.
3. Analyses: `python src/02_eda_reviews.py`, `src/03_eda_games.py`, `src/04_eda_joined.py`. Figures go to `reports/stage1/figures/`, numbers to `reports/stage1/results/`.

The prepared files (about 15 MB in total) are also in the repository, so the analyses can be run without downloading the data.

4. Exploratory analysis – dataset 1 (reviews)

Methods: descriptive statistics, 95% Wilson confidence intervals for proportions, the χ^2 test of independence, the Mann-Whitney U test (with the rank-biserial effect size r), and vocabulary comparison with log-odds and an informative Dirichlet prior (Monroe et al., 2008). All p-values reported as "$p < 0.001$" were in fact smaller than 10^{-20}.

4.1 Data quality and cleaning

- There are no missing values in the key columns (text, rating, game). The year is missing from the date of 10,119 reviews: Steam does not show the year for posts from the current year, and the data has no review dated 2016, so we assume these reviews are from 2016 (the year the data was collected).
- 38 reviews are empty and 874 are repeated (user, game) pairs – all with identical text, i.e. copies created during data collection. After removing them, **58,393 reviews** remain from 25,444 users about 3,682 games.
- Texts can be very short: 16.6% of reviews have at most 3 words, and the most frequent repeated texts are "good game" (101 times), "good" (100) and "10/10" (77). Some swear words are masked by Steam with "♥♥♥♥".
- Language (a simple heuristic): 69.9% of reviews are in English, 23.0% are too short to tell, 5.9% are in other languages written in the Latin alphabet, and 1.3% use Cyrillic or Chinese, Japanese and Korean characters.

4.2 Target variable and changes over time



Figure 1. Number of "recommended" and "not recommended" reviews, number of reviews per year and share of "recommended" per year (2016 – reviews without a year in the date).*

- **The classes are imbalanced:** 88.5% of reviews recommend the game (95% CI: 88.2–88.7%), only 11.5% (6,736) do not. Models therefore need to be judged by F1 and recall for the "not recommended" class, not only by accuracy.
- **The share of "recommended" falls over time** from 98.5% in 2010 to 81.9% in 2016 ($\chi^2 = 1219.7$, df = 6, $p < 0.001$). Most reviews come from 2014–2015.

4.3 Review length

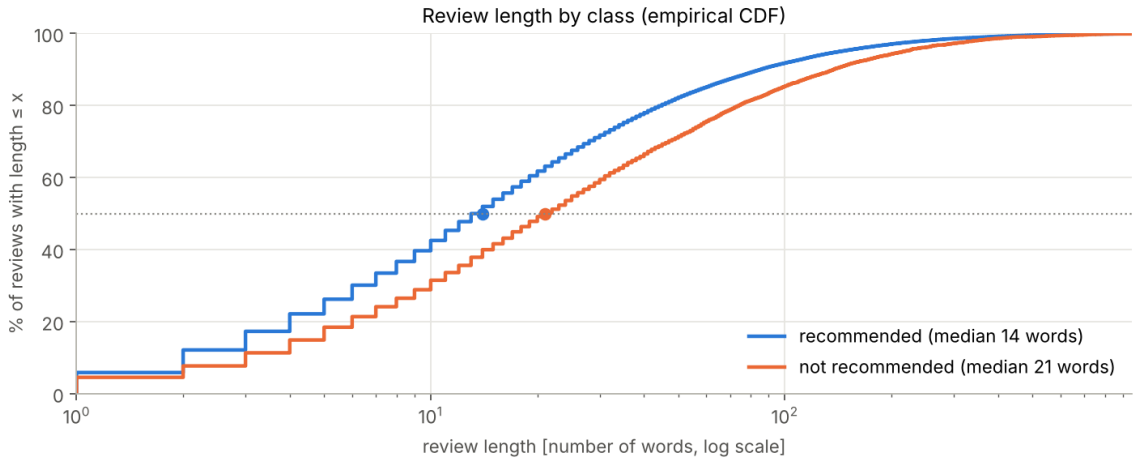


Figure 2. Empirical cumulative distribution of review length: the share of reviews with at most x words.

The median length is 14 words, the mean 38.7 and the maximum 2,667 words. **Negative reviews are longer:** a median of 21 words versus 14 in positive ones (Mann-Whitney U test: $p < 0.001$, $r = 0.17$ – a small effect). A player who does not recommend a game more often explains why. Negative reviews also receive more "helpful" votes: a median of 3 versus 0 ($p < 0.001$, $r = 0.36$).

4.4 Games in the review dataset

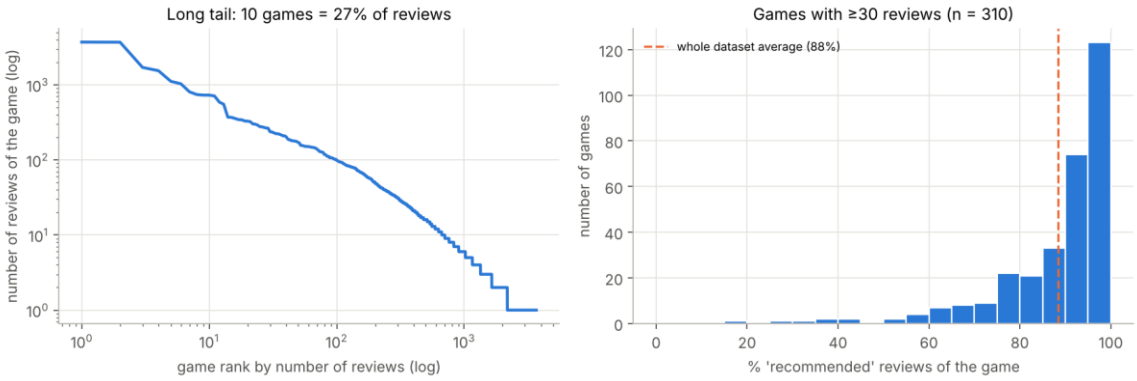


Figure 3. Left: number of reviews of a game against its rank (log-log scale). Right: share of "recommended" for games with at least 30 reviews.

The distribution is very skewed: the median is 2 reviews per game, 1,504 games have only one review, and the 10 most popular games collect 27% of all reviews (the most: Counter-Strike: Global Offensive,

now Counter-Strike 2, and Team Fortress 2 – about 3,700 each). Among the 310 games with at least 30 reviews, the median share of "recommended" is 93.1%, but some games are below 30%.

4.5 Vocabulary

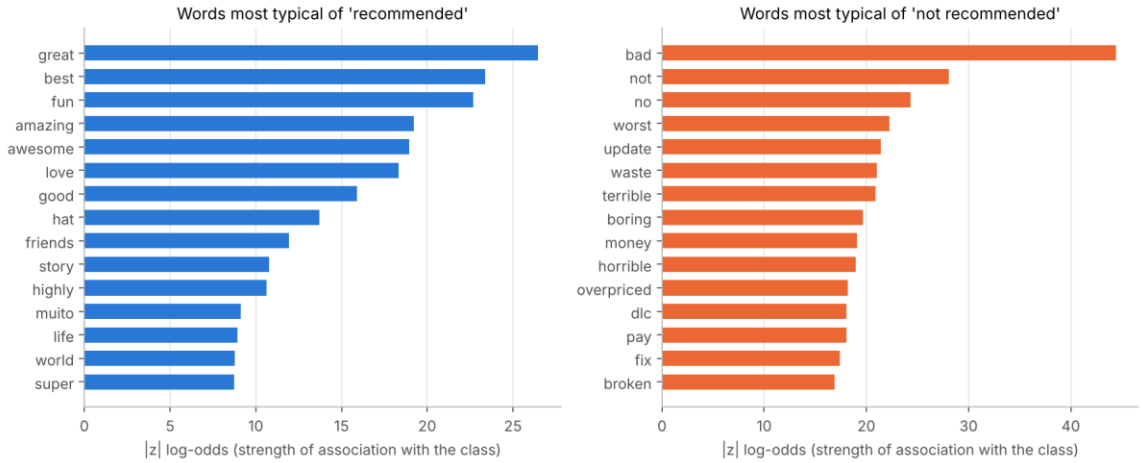


Figure 4. Words most strongly associated with each class (log-odds z-score; function words other than negations, and the words "game" and "play", are excluded).

The corpus has 2.1 million words and 68,152 distinct words, of which 39,566 occur only once. Words typical of positive reviews are general judgements (great, best, fun, amazing) as well as "friends" and "story". In negative reviews, besides judgements (bad, worst, terrible) and negations (not, no), there are **concrete reasons**: updates and bugs (update, fix, broken) and money (money, overpriced, dlc, pay). These are the topics a topic model should find in the next stages.

5. Exploratory analysis – dataset 2 (games)

Methods: descriptive statistics, analysis of missing values and zeros, Spearman rank correlation, and statistical tests: the Mann-Whitney U test, the Kruskal-Wallis test and the χ^2 test. The share of positive ratings is computed only for games with at least 50 ratings, because with a handful of ratings it is very unstable:

```
\text{positive share} = \frac{\text{positive}}{\text{positive} + \text{negative}}
```

"Success" means at least 80% positive ratings (on Steam: "Very Positive").

5.1 Data quality

- There are no duplicate game IDs and no missing numerical values, but **zero often means "no information"**: the Metacritic score is 0 for 96.6% of games, the Metacritic user score for 99.97%, the required age for 99.0% and the average playtime for 79.0%. These columns cannot be treated as ordinary numbers.
- 33.2% of games have no rating at all, and 23,033 games have 0 owners and 0 ratings: 35% of them have "Playtest" in the name (test versions) and 45% were released in 2025. The number of owners is only given as a range (e.g. "0 – 20000").
- 8,419 games have no genre. The tags column is empty after reading the Parquet file – in stage 1 we use genres and categories, and we will look into reading the tags in the next stage.

5.2 Distributions

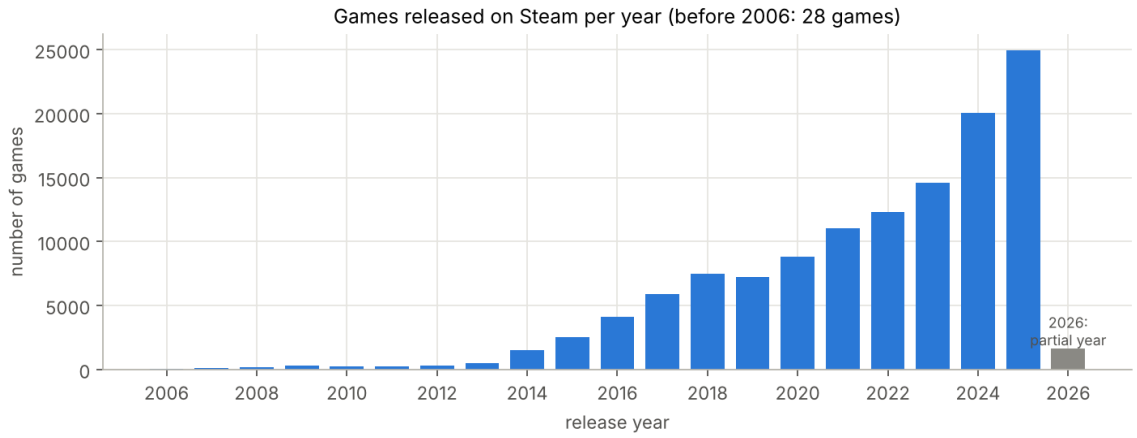


Figure 5. Number of games released on Steam per year (2026 – data collected during the year).

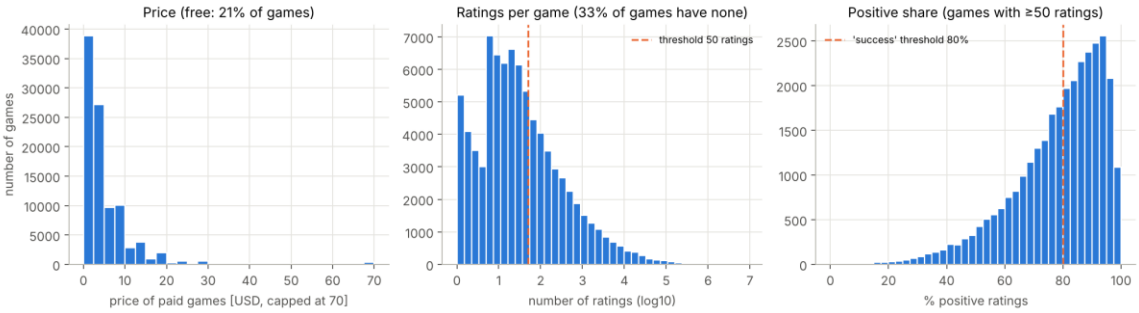


Figure 6. Distribution of the price of paid games, the number of ratings (log scale) and the share of positive ratings (games with at least 50 ratings).

- The number of released games is growing fast: from 467 in 2013 to 24,973 in 2025.
- 21.3% of games are free. Paid games are usually cheap: the median is 3.59 USD and 90% cost at most 11.99 USD.
- The number of ratings is extremely skewed: median 7, 90th percentile 366, maximum 8.8 million. 30,621 games (24.7%) meet the 50-rating threshold, and they are the basis of the further rating analyses.
- The median share of positive ratings is 82.0%, and 55.1% of games meet the “success” threshold. The distribution is shifted towards high ratings.
- Genres: most often Indie (81,746 games), Casual, Action and Adventure; a game can have several genres. 99.96% of games run on Windows, 17.3% on macOS and 12.8% on Linux.

5.3 Relationships and statistical tests

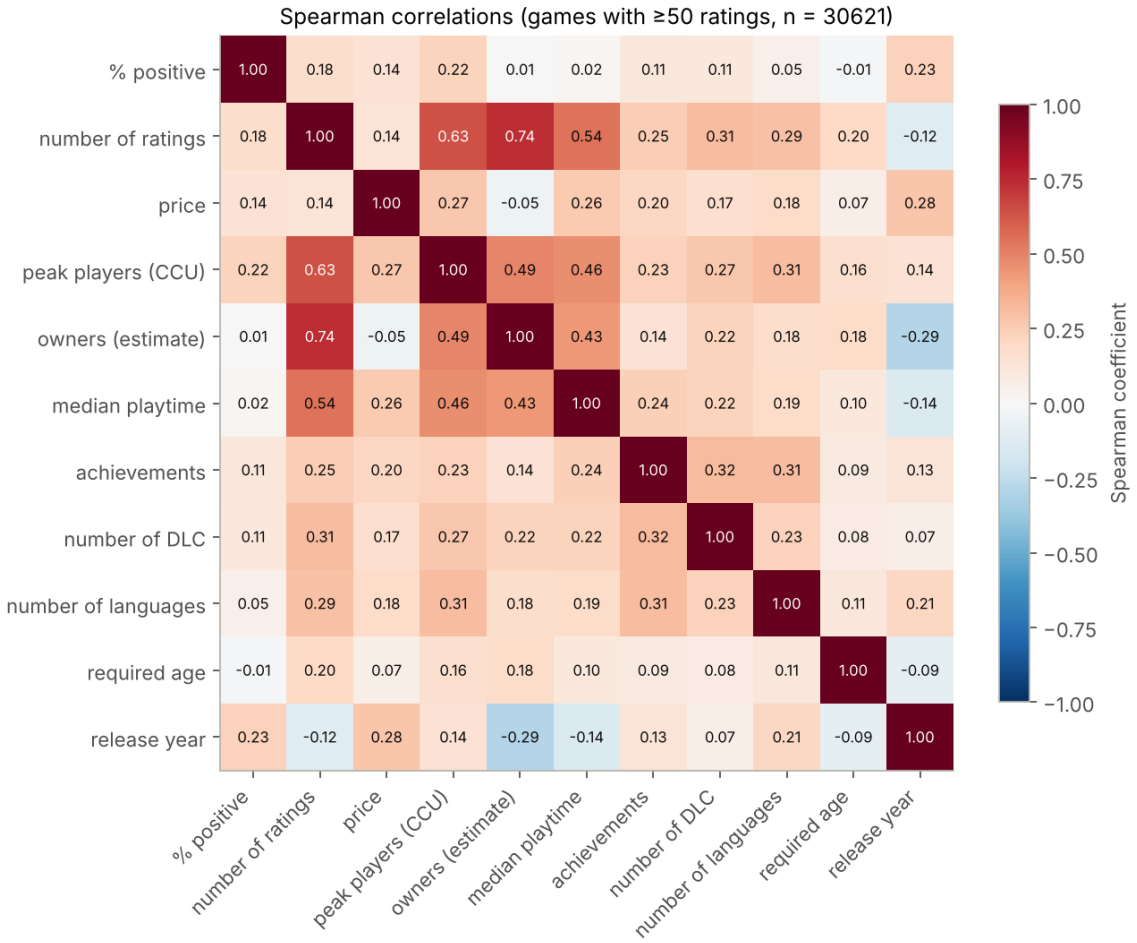


Figure 7. Spearman correlations between features of games with at least 50 ratings.

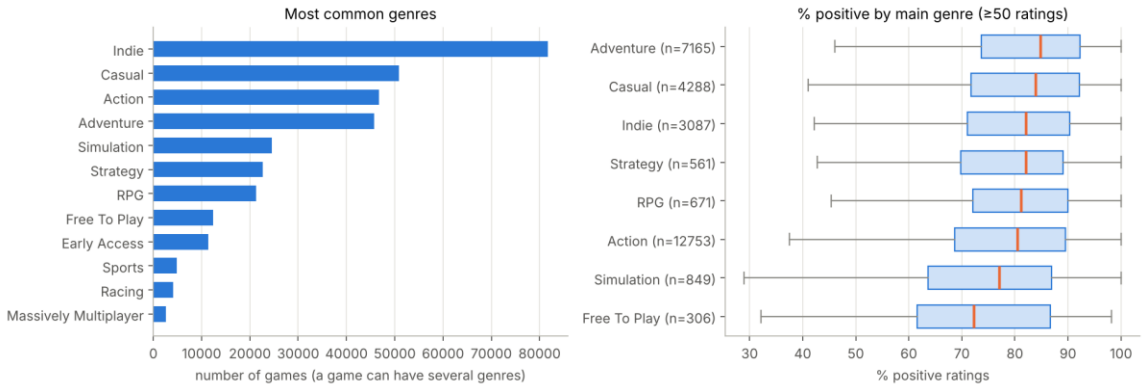


Figure 8. Most common genres and the share of positive ratings by main genre (the first one listed).

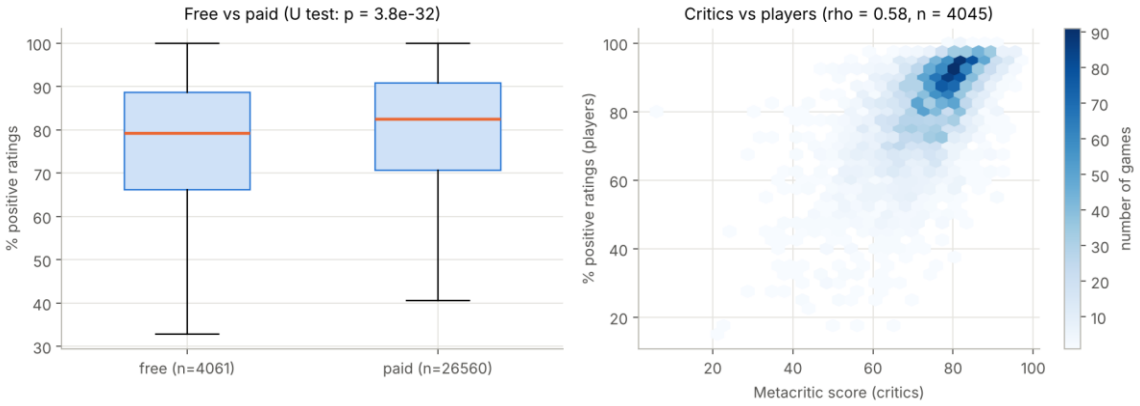


Figure 9. Share of positive ratings of free and paid games, and critics' score (Metacritic) against players' ratings.

- **Single features are only weakly related to players' ratings.** The strongest Spearman correlations with the positive share are release year (ρ = 0.23), peak number of players (0.22), number of ratings (0.18) and price (0.14). The number of owners, playtime and required age show almost no relationship ($|\rho| \leq 0.02$). The features are, however, strongly correlated with each other: the number of ratings with the number of owners (0.74) and the player peak (0.63). These measure a game's popularity, not its quality.
- **Free games are rated lower:** a median of 79.2% versus 82.4% for paid games (U test: $p < 0.001$, $r = 0.11$). 48.1% of free games and 56.2% of paid games reach "success" ($\chi^2 = 93.6$, $p < 0.001$).
- **The genre matters** (Kruskal-Wallis for the 8 most common main genres: $H = 549.0$, $p < 0.001$): adventure games (median 84.9%) and casual games (83.9%) are rated highest, simulations (77.1%) and Free To Play games (72.3%) lowest.
- **Critics and players rate similarly, but not identically:** the correlation between the Metacritic score and the players' positive share is ρ = 0.58 (4,045 games with a Metacritic score, $p < 0.001$).

6. Joining the datasets

Reviews are joined with games on the game ID. The games dataset contains 2,750 of the 3,682 games from the reviews (74.7%); they account for 88.4% of the reviews. The remaining 932 games (6,801 reviews) are not in dataset 2 – they may be titles removed from the store or items of a different type; we will check this in the next stage.

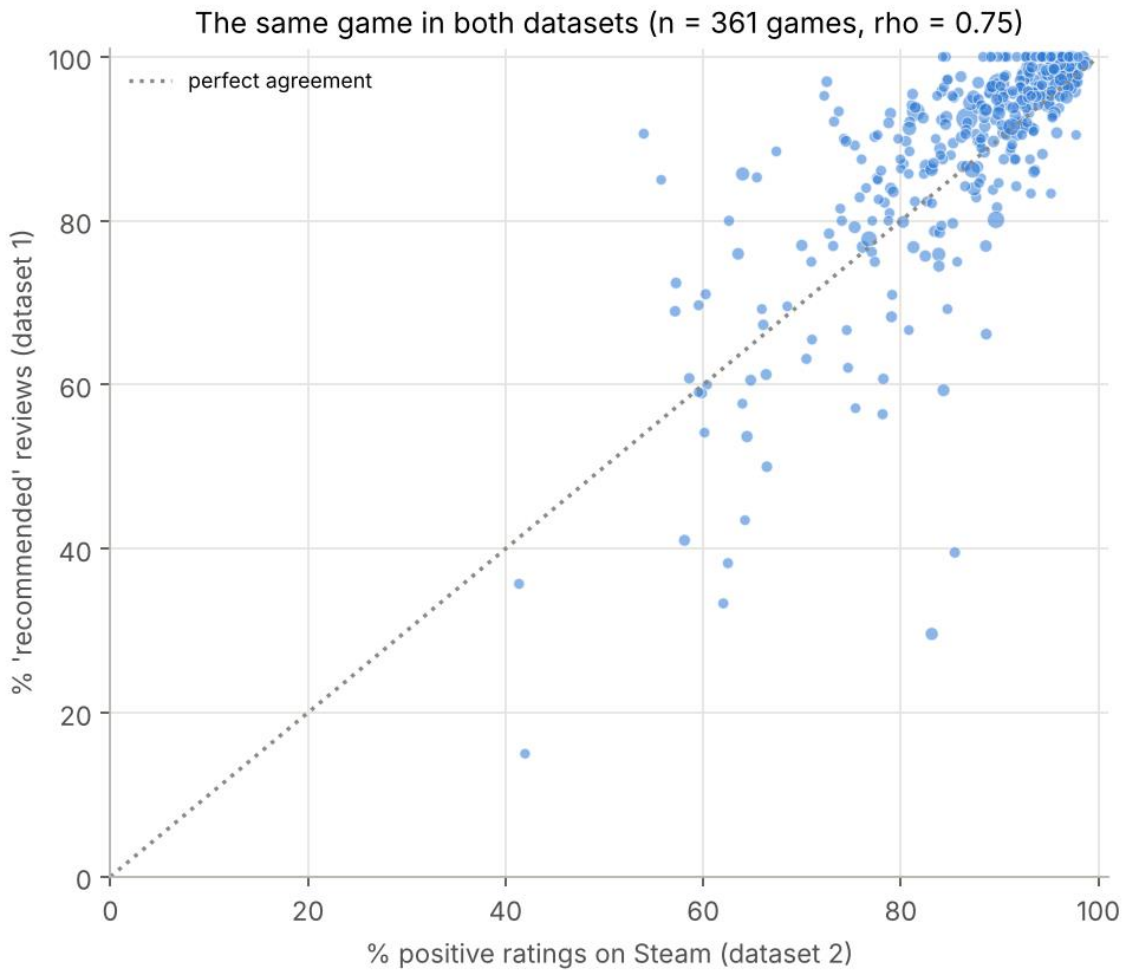


Figure 10. Games with at least 20 reviews in dataset 1 and at least 50 ratings in dataset 2: share of positive ratings on Steam (X axis) against the share of "recommended" reviews (Y axis). Point size = number of reviews.

- **The two datasets agree:** for 361 games the Spearman correlation between the share of "recommended" reviews and the share of positive ratings on Steam is $\rho = 0.75$ ($p < 0.001$), the mean difference is +2.2 percentage points and the median absolute difference 4.5 pp.
- **The largest disagreements make sense:** No Man's Sky is recommended in only 29.6% of its 179 reviews from 2016 (the year of its unsuccessful launch), while today it has 83.1% positive ratings on Steam, because it was developed and improved over the following years. Similarly Call of Duty: Black Ops III (59.3% versus 84.3%). The reviews describe the opinion at the time of writing, dataset 2 the current state.
- **At the review level, the game's price does not matter:** reviews of free and paid games recommend them equally often (90.2% versus 89.8%, $\chi^2 = 1.6$, $p = 0.21$). This differs from the result for all games on Steam, where free games do worse. Two possible reasons: the review dataset is dominated by very popular free games (Team Fortress 2, Dota 2), and the price comes from dataset 2, i.e. the current state – e.g. Counter-Strike: Global Offensive (now Counter-Strike 2; 3,704 reviews from 2012–2016) is free today, but it was paid when the reviews were written.

- **What does matter is whether the review talks about money:** 9.1% of reviews mention price (price, money, worth, refund, overpriced etc.), and these reviews recommend the game in 80.7% of cases versus 90.9% for the rest ($\chi^2 = 482.3$, $p < 0.001$). Players of paid games write about price more often (10.6%) than players of free games (6.4%).
- The share of "recommended" also differs between main genres ($\chi^2 = 143.9$, $df = 7$, $p < 0.001$): from 93.8% for Casual games to 80.5% for Free To Play (figure 11).

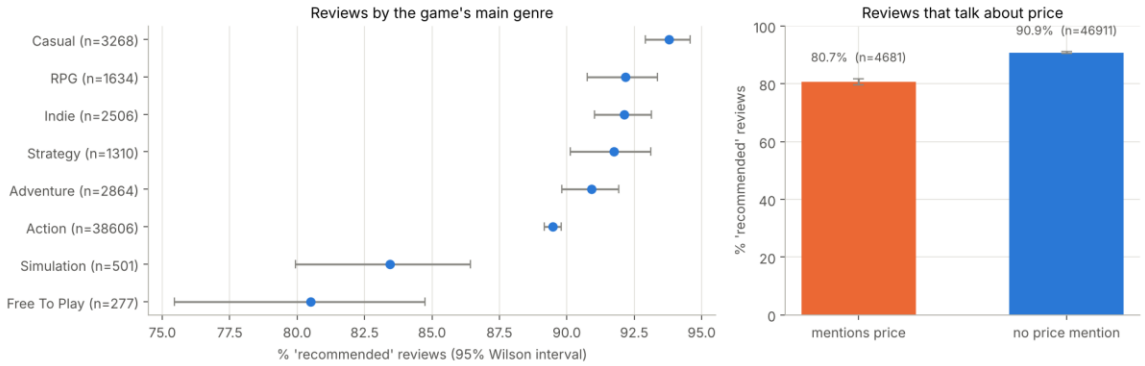


Figure 11. Share of "recommended" reviews by the game's main genre and by whether the review mentions price (95% Wilson intervals).

7. Research questions

Main question: what makes players rate a Steam game well, and can it be predicted from the review text and from the game's features?

- **Q1. Review text (supervised learning, NLP).** Can we predict from the text alone whether a player recommends the game? How does a classical model (TF-IDF + logistic regression) compare with a transformer-based model (e.g. DistilBERT), especially for the rarer "not recommended" class (11.5%) and for very short reviews (16.6% have at most 3 words)?
- **Q2. Review topics (unsupervised learning, NLP).** What do players write about, and which topics dominate negative reviews? The vocabulary analysis suggests: price and DLC, updates and bugs, playing with friends, story.
- **Q3. Game features and rating (supervised learning).** To what extent do game features – price, genre, platforms, number of languages, achievements, DLC, release year – predict the share of positive ratings or "success" (at least 80%)? Which features matter most? The weak correlations of single features ($|\rho| \leq 0.23$) suggest that their combination matters rather than any one feature.
- **Q4. Game segments (unsupervised learning).** Do games form natural groups, e.g. free online games, cheap indie games, big productions, and do the groups differ in players' ratings?
- **Q5. Text versus numbers (both datasets).** Does a rating derived from review text agree with the share of positive ratings on Steam, and when do they diverge – because of time (No Man's Sky), price or genre? Is the drop in the share of "recommended" from 98.5% in 2010 to 81.9% in 2016 caused by a change in the kind of games being reviewed, or by harsher ratings?

Planned methods in the next stages: supervised learning – logistic regression and random forest or gradient boosting; unsupervised learning – k-means with PCA for games and topic modelling (LDA) for

reviews; a transformer model (DistilBERT) for review classification; a simple Gradio interface for exploring the results (scoring any review text and predicting the rating for given game features).

8. Limitations

- Dataset 1 only covers reviews by users from Australia from 2010–2016, so it is not representative of all players or of today's Steam.
- The datasets come from different years: reviews from 2010–2016, game ratings as of 2026. Some differences between the datasets are due to time, not errors.
- Language detection is a simplified heuristic, and the number of owners is only given as a range.
- The share of positive ratings is computed only for games with at least 50 ratings. That is 24.7% of all games, mostly the more popular titles.

9. Repository

Code, prepared data, figures, numerical results and reports: https://github.com/ProstoKery/steam-ml-project. The `README.md` file explains how to reproduce the results.